

(2)

:

$$\begin{aligned}
 & \text{1 } \Omega \subset \mathbb{R}^n \qquad f: \Omega \rightarrow \mathbb{R} \\
 & \Omega \subset \mathbb{R}^n \\
 & \frac{\partial f}{\partial x_i} \quad (1 \leq i \leq n) \qquad \frac{\partial f}{\partial x_i} \quad j \\
 & \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) \\
 & \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \partial_{ji} f \qquad i = j \\
 & \frac{\partial^2 f}{\partial x_i^2}
 \end{aligned}$$

⋮
1 ?
 $\forall (xy) \in \mathbb{R}^2$

$$f(xy) = \begin{cases} \frac{(x^2-y^2)xy}{x^2+y^2} & (xy) \neq (00) \\ 0 & (xy) = (00) \end{cases}$$
$$\frac{\partial^2 f}{\partial y \partial x}(00) \neq \frac{\partial^2 f}{\partial x \partial y}(00)$$

⋮
 ?

$$\begin{array}{l} 1 \quad \Omega \subset \mathbb{R}^n \quad f: \Omega \rightarrow \mathbb{R} \\ \Omega \quad \frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (i \neq j) \\ \quad \quad X_0 \in \Omega \end{array}$$

$$\frac{\partial^2 f}{\textcolor{red}{\partial x_j} \textcolor{blue}{\partial x_i}}(X_0) = \frac{\partial^2 f}{\textcolor{blue}{\partial x_i} \textcolor{red}{\partial x_j}}(X_0)$$

$$2 \quad \Omega \subset \mathbb{R}^n \quad k \geq 0$$

$$\mathcal{C}^{(0)}(\Omega) = \mathcal{C}(\Omega) \quad f \in \mathcal{C}^{(k)}(\Omega)$$

$$f \in \mathcal{C}^{(k)}(\Omega) \quad k$$

$$2 \quad k \geq 2 \quad f \in \mathcal{C}^{(k)}(\Omega)$$

$$1 \leq r \leq k \quad f \in \mathcal{C}^{(r)}(\Omega) \quad f$$

$$r$$

$$\mathcal{C}^{(k)}(\Omega) = \{f \mid f \in \mathcal{C}^{(k)}(\Omega)\}$$

5in

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:

$$\mathbb{R}^2 \quad D \quad z = f(uv) \quad \mathbb{R}^2 \quad D_2 \quad :$$

$$u = u(xy); \quad v = v(xy)$$

$$z = F(xy) = f(u(xy)v(xy)) \quad (xy) \in D$$

$$\text{(\hspace{0.05cm} \textcolor{blue}{x} \hspace{0.05cm})} \qquad u = u(xy); \quad v = v(xy) \quad (xy) \in D \qquad z = f(uv) \quad (uv) = (u(xy)v(xy)) \qquad :$$

$$(xy)$$

$$z = F(xy) = f(u(xy)v(xy))$$

$$\frac{\partial z}{\textcolor{red}{\partial x}} = \frac{\partial z}{\partial u} \cdot \frac{\textcolor{blue}{\partial u}}{\textcolor{red}{\partial x}} + \frac{\partial z}{\partial v} \cdot \frac{\textcolor{blue}{\partial v}}{\textcolor{red}{\partial x}};$$

$$\frac{\partial z}{\textcolor{red}{\partial y}} = \frac{\partial z}{\partial u} \cdot \frac{\textcolor{blue}{\partial u}}{\textcolor{red}{\partial y}} + \frac{\partial z}{\partial v} \cdot \frac{\textcolor{blue}{\partial v}}{\textcolor{red}{\partial y}};$$

$$x+\Delta x-uv\quad:$$

$$\Delta_x u = u(x+\Delta xy) - u(xy);$$

$$\Delta_x v = v(x+\Delta xy) - v(xy)$$

$$z=f(uv)\quad (uv)$$

$$\Delta z := \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + o(\sqrt{\Delta u^2 + \Delta v^2})$$

:
:
:

$$\Delta_x z := \frac{\partial z}{\partial u} \Delta_x u + \frac{\partial z}{\partial v} \Delta_x v + o(\sqrt{\Delta_x u^2 + \Delta_x v^2})$$

$$\frac{\Delta_x z}{\Delta x} = \frac{\partial z}{\partial u} \cdot \frac{\Delta_x u}{\Delta x} + \frac{\partial z}{\partial v} \cdot \frac{\Delta_x v}{\Delta x} + \frac{o(\sqrt{\Delta_x u^2 + \Delta_x v^2})}{\Delta x}$$

$$\sqrt{\Delta_x u^2 + \Delta_x v^2} := \rho$$

$$\Delta x \rightarrow 0 \quad :$$

$$\frac{\partial u \partial v}{\partial x \partial x} \qquad \Delta x \rightarrow 0$$

$$\begin{aligned} \Delta_x u &= u(x + \Delta xy) - u(xy) \rightarrow 0; \\ \Delta_x v &= v(x + \Delta xy) - v(xy) \rightarrow 0; \end{aligned}$$

$$\begin{aligned} \frac{\Delta_x u}{\Delta x} &= \frac{u(x + \Delta xy) - u(xy)}{\Delta x} \rightarrow \frac{\partial u}{\partial x}; \\ \frac{\Delta_x v}{\Delta x} &= \frac{v(x + \Delta xy) - v(xy)}{\Delta x} \rightarrow \frac{\partial v}{\partial x}; \end{aligned}$$

:

$$\rho = \sqrt{\Delta_x u^2 + \Delta_x v^2} \rightarrow 0;$$

$$\frac{o(\rho)}{\rho} = \frac{o(\sqrt{\Delta_x u^2 + \Delta_x v^2})}{\sqrt{\Delta_x u^2 + \Delta_x v^2}} \rightarrow 0$$

: $\Delta x \rightarrow 0 \qquad \rho = 0 \qquad \frac{o(\rho)}{\Delta x} = 0;$

$$\rho \neq 0$$

$$\frac{o(\rho)}{\rho} \cdot \sqrt{\left(\frac{\Delta_x u}{\Delta x}\right)^2 + \left(\frac{\Delta_x v}{\Delta x}\right)^2} \cdot \frac{|\Delta x|}{\Delta x} \rightarrow 0$$

$$\Delta x \rightarrow 0$$
$$\vdots$$

$$\begin{aligned}\frac{\Delta_x z}{\Delta x} &= \frac{\partial z}{\partial u} \cdot \frac{\Delta_x u}{\Delta x} + \frac{\partial z}{\partial v} \cdot \frac{\Delta_x v}{\Delta x} + \frac{o(\sqrt{\Delta_x u^2 + \Delta_x v^2})}{\Delta x} \\ \frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}; \\ \frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y};\end{aligned}$$

- $z = f(uv) = u^2v - uv^2 \quad u = x \sin y \quad v = x \cos y \quad \frac{\partial z}{\partial x} \quad :$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} \\ &= \frac{\partial f}{\partial u} \cdot \frac{\partial (x \sin y)}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial (x \cos y)}{\partial x} \\ &= (2uv - v^2) \sin y + (u^2 - 2uv) \cos y \\ &= (2x^2 \sin y \cos y - x^2 \cos^2 y) \sin y \\ &\quad + (x^2 \sin^2 y - 2x^2 \sin y \cos y) \cos y \\ &= \frac{3}{2}x^2(\sin y - \cos y) \sin(2y) \end{aligned}$$

$$\begin{array}{ll} 2 & z = f(xy^2 - y^2) \quad f \quad \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \\ : & z = f(uv); \quad u = xy; v = x^2 - y^2 \end{array}$$

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial f \partial u}{\partial u \partial x} + \frac{\partial f \partial v}{\partial v \partial x} \\ &= \partial_1 f(xyx^2 - y^2) \frac{\partial(xy)}{\partial x} \\ &\quad + \partial_2 f(xyx^2 - y^2) \frac{\partial(x^2 - y^2)}{\partial x} \\ &= y\partial_1 f(xyx^2 - y^2) + 2x\partial_2 f(xyx^2 - y^2)\end{aligned}$$

$$\begin{aligned}
 \frac{\partial z}{\partial y} &= \partial_1 f(xy^2 - y^2) \frac{\partial(xy)}{\partial y} \\
 &\quad + \partial_2 f(xy^2 - y^2) \frac{\partial(x^2 - y^2)}{\partial y} \\
 &= x \partial_1 f(xy^2 - y^2) - 2y \partial_2 f(xy^2 - y^2)
 \end{aligned}$$

3 $z = \frac{y}{x} + xyf(\frac{y}{x})$ $f'(\frac{y}{x}) = \frac{\partial z}{\partial x}$
:

$$\begin{aligned}\frac{\partial z}{\partial x} &= -\frac{y}{x^2} + yf'(\frac{y}{x}) + xy \cdot f'(\frac{y}{x}) \cdot (-\frac{y}{x^2}) \\ &= -\frac{y}{x^2} + yf'(\frac{y}{x}) - \frac{y^2}{x}f'(\frac{y}{x})\end{aligned}$$

4

(Laplace)

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad r = \sqrt{x^2 + y^2 + z^2} \quad : \quad \mathbb{R}^3 \setminus \{(000)\}$$

:

$\mathbb{R}^3 \setminus \{(000)\}$

$$\Delta \left(\frac{1}{r} \right) = \frac{\partial^2 \left(\frac{1}{r} \right)}{\partial x^2} + \frac{\partial^2 \left(\frac{1}{r} \right)}{\partial y^2} + \frac{\partial^2 \left(\frac{1}{r} \right)}{\partial z^2} = 0$$

$$\frac{\partial}{\partial x} \left(\frac{1}{r} \right) = -\frac{1}{r^2} \frac{\partial r}{\partial x} = -\frac{1}{r^2} \frac{x}{\sqrt{x^2 + y^2 + z^2}} = -\frac{x}{r^3}$$

$$\frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) = -\frac{1}{r^3} + \frac{3x \partial r}{r^4 \partial x} = -\frac{1}{r^3} + \frac{3x^2}{r^5}$$

$$\begin{aligned}
 \Delta\left(\frac{1}{r}\right) &= \left(-\frac{1}{r^3} + \frac{3x^2}{r^5}\right) + \left(-\frac{1}{r^3} + \frac{3y^2}{r^5}\right) \\
 &\quad + \left(-\frac{1}{r^3} + \frac{3z^2}{r^5}\right) \\
 &= -\frac{3}{r^3} + \frac{3(x^2 + y^2 + z^2)}{r^5} = 0
 \end{aligned}$$

- :

$\mathcal{C}^{(k)}(\Omega)$
- a

$\mathcal{C}^{(k)}(\Omega) \ (k \geq 0 \)$
- b

$f \in \mathcal{C}^{(k)}(\Omega) \ \Omega \ \text{is } k \text{-times differentiable}$
- c

$k \geq 2 \ \ f \in \mathcal{C}^{(k)}(\Omega) \ \ f \in \mathcal{C}^{(r)}(\Omega) \ \ f \text{ is } r \text{-times differentiable}$

5in

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$$\textcolor{red}{:} \quad F(xy) = 0? \qquad F(xy) = 0 \quad y = y(x)?$$

$\textcolor{blue}{:}$ $\textcolor{blue}{:}$

$$\begin{aligned} F(xy) &= ax + by + c \\ F(xy) = 0 \quad y \quad \frac{\partial F}{\partial y} &= b \neq 0 \\ y &= -\frac{1}{b}(ax + c) \end{aligned}$$

:

$$F(xy) := x^2 + y^2 - 1 = 0$$

$$y = \pm \sqrt{1 - x^2}$$

1

$y > 0 \quad y = \sqrt{1 - x^2} \quad \frac{\partial F}{\partial y} = 2y > 0$

2

$y < 0 \quad y = -\sqrt{1 - x^2} \quad \frac{\partial F}{\partial y} = 2y < 0$

3

(10)

$y \quad \frac{\partial F}{\partial y}(10) = 0$

:

$F(xy) = 0 \quad y = y(x) \quad \frac{\partial F}{\partial y} \quad ?$

$$\mathbb{R}^2$$

$$\textcolor{blue}{1} \quad X_0 = (x_0y_0) \in \mathbb{R}^2 \; r > 0 \qquad F \colon B(X_0r) \rightarrow \mathbb{R} \; \mathcal{C}^{(1)}$$

$$F(x_0y_0) = 0 \quad \textcolor{red}{\frac{\partial F}{\partial y}}(x_0y_0) \neq 0$$

$$\exists \delta \eta > 0$$

$$\forall x(x_0\delta) \qquad \textcolor{red}{(\exists!)} \; y \in B(y_0\eta) \quad F(xy) = 0$$

$$B(x_0\delta) \times B(y_0\eta) \subset B(X_0r)$$

$$f(x) = y$$

$$\mathcal{C}^{(1)} \qquad \forall x \in B(x_0\delta)$$

$$f\colon B(x_0\delta)\rightarrow B(y_0\eta)$$

$$\textcolor{red}{f}(x) = -\frac{\frac{\partial F}{\partial x}(x\textcolor{red}{f}(x))}{\frac{\partial F}{\partial y}(x\textcolor{red}{f}(x))}$$

$$\begin{array}{l} \vdots \quad \frac{\partial F}{\partial y}(x_0 y_0) > 0 \quad -F \\ \vdots \quad \frac{\partial F}{\partial y} \quad \exists \eta > 0 \quad \forall (xy) \in B(X_0 \sqrt{2} \eta) \subsetneq B(X_0 r) \quad \frac{\partial F}{\partial y}(xy) > 0 \quad \forall (xy) \in B(X_0 \sqrt{2} \eta) \quad g_x(y) = F(xy) \quad x \in [x_0 - \eta x_0 + \eta] \quad g_x \quad [y_0 - \eta y_0 + \eta] \quad g'_{x_0}(y) = \frac{\partial F}{\partial y}(x_0 y) > 0 \quad g_{x_0} \end{array}$$

$$g_{x_0}(y_0) = 0$$

$$\begin{aligned} F(x_0y_0 - \eta) &= g_{x_0}(y_0 - \eta) < g_{x_0}(y_0) = 0 \\ &< g_{x_0}(y_0 + \eta) = F(x_0y_0 + \eta) \end{aligned}$$

$$F \qquad \qquad \exists \delta \in (0\eta) \quad \forall x \in (x_0 - \delta x_0 + \delta)$$

$$\begin{aligned} g_x(y_0 - \eta) &= F(xy_0 - \eta) < 0 \\ g_x(y_0 + \eta) &= F(xy_0 + \eta) > 0 \end{aligned}$$

$$\forall y \in [y_0 - \eta y_0 + \eta] \quad g'_x(y) = \frac{\partial F}{\partial y}(xy) > 0 \quad g_x \left[y_0 - \eta y_0 + \eta \right]$$

$$\exists! y \in (y_0 - \eta y_0 + \eta) \quad F(xy) = g_x(y) = 0 \quad f(x) = y \quad f$$

$$: \qquad \forall \varepsilon \in (0\eta) \, \exists \delta' \in (0\varepsilon) \, \forall x \in B(x_0\delta') \, \exists ! y(y_0\varepsilon) \, F(xy) = 0 \quad y = f(x) \quad |x - x_0| < \delta' \qquad |f(x) - f(x_0)| < \varepsilon \quad f \, x_0 \qquad x_1(x_0\delta) \, y_1 = f(x_1) \, F(x_1y_1) = 0 \, (x_1y_1) \in B((x_0y_0)\eta) \, \frac{\partial F}{\partial y}(x_1y_1) > 0$$

$$\delta_1 \in (0\delta)\eta_1 \in (0\eta) \quad x_1 \quad g: B(x_1\delta_1) \rightarrow B(y_1\eta_1) \quad F(xg(x)) = 0 \quad B(x_1\delta_1)(x_0\delta)$$

$$\begin{array}{l} \forall x \in B(x_1, \delta_1) \quad f(x) = g(x) \quad f(x_1) \\ \vdots \quad x(x_0, \delta) \quad h \in \mathbb{R} \quad x + h(x_0, \delta) \quad y = f(x) \quad \Delta y = f(x + h) - f(x) \quad \text{Lagrange} \quad \exists \theta_1, \theta_2 \in (0, 1) \end{array}$$

$$\begin{aligned} 0 &= F(x + hy + \Delta y) - F(xy) \\ &= \left(F(x + hy + \Delta y) - \textcolor{red}{F(xy + \Delta y)} \right) \\ &\quad + \left(\textcolor{red}{F(xy + \Delta y)} - F(xy) \right) \\ &= \frac{\partial F}{\partial x}(x + \theta_1 hy + \Delta y)h + \frac{\partial F}{\partial y}(xy + \theta_2 \Delta y)\Delta y \end{aligned}$$

:

$$\frac{\partial F \partial F}{\partial x \partial y}$$

$$0 = \frac{\partial F}{\partial x}(x + \theta_1 h y + \Delta y) \textcolor{red}{h} + \frac{\partial F}{\partial y}(x y + \theta_2 \Delta y) \textcolor{red}{\Delta y}$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{\textcolor{red}{h} \rightarrow 0} \frac{\textcolor{red}{\Delta y}}{\textcolor{red}{h}} \\ &= - \lim_{h \rightarrow 0} \frac{\frac{\partial F}{\partial x}(x + \theta_1 h y + \Delta y)}{\frac{\partial F}{\partial y}(x y + \theta_2 \Delta y)} \\ &= \end{aligned}$$

f' f

$$f'(x) = -\frac{\frac{\partial F}{\partial x}(xy)}{\frac{\partial F}{\partial y}(xy)} = -\frac{\frac{\partial F}{\partial x}(xf(x))}{\frac{\partial F}{\partial y}(xf(x))}$$

$$\textcolor{blue}{2} \quad X_0 \in \mathbb{R}^n \quad y_0 \in \mathbb{R} \quad r > 0 \\ F \colon B((X_0y_0)r) \rightarrow \mathbb{R} \quad \mathcal{C}^{(1)} \quad F(X_0y_0) = 0 \quad \frac{\partial F}{\partial y}(X_0y_0) \neq 0 \quad \exists \delta \eta > 0$$

$$\forall X(X_0\delta) \quad \exists ! y(y_0\eta) \; F(Xy) = 0 \quad f(X) = y \quad f \colon B(X_0\delta)(y_0\eta) \quad \mathcal{C}^{(1)} \quad \forall X \in B(X_0\delta) \quad 1 \leq i \leq n$$

$$B(X_0\delta) \times B(y_0\eta) \subset B\Big((X_0y_0);r\Big)$$

$$\frac{\partial f}{\partial x_i}(\textcolor{red}{X}) = -\frac{\frac{\partial F}{\partial x_i}(Xf(X))}{\frac{\partial F}{\partial y}(Xf(X))}$$

$$x_i$$

$$F(x_1 \dots x_n f(x_1 \dots x_n)) = 0$$

$$\frac{\partial F}{\partial x_i}(Xf(X)) + \frac{\partial F}{\partial y}(Xf(X))\frac{\partial f}{\partial x_i}(X) = 0$$

$$\frac{\partial f}{\partial x_i}(X) = -\frac{\frac{\partial F}{\partial x_i}(Xf(X))}{\frac{\partial F}{\partial y}(Xf(X))}$$

:

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m \qquad \mathbf{e}_1 \dots \mathbf{e}_n \in \mathbb{R}^n \qquad \mathbf{f}_1 \dots \mathbf{f}_m \in \mathbb{R}^m$$

$$a_{ij} = A\mathbf{e}_j \cdot \mathbf{f}_i \quad (1 \leq i \leq m, 1 \leq j \leq n) \qquad A\mathbf{e}_j = \sum_{i=1}^m a_{ij}\mathbf{f}_i \qquad X = \sum_{j=1}^n x_j\mathbf{e}_j$$

$$AX = \sum_{j=1}^n x_j A\mathbf{e}_j = \sum_{j=1}^n x_j \sum_{i=1}^m a_{ij}\mathbf{f}_i = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}x_j \right) \mathbf{f}_i$$

$$(AX)_i = (AX) \cdot \mathbf{f}_i = \sum_{j=1}^n a_{ij}x_j \qquad A = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$$

$$\textcolor{blue}{1} \quad X_0 = (x_1^{(0)} \dots x_n^{(0)}) \in \mathbb{R}^n \; r > 0 \; \vec{f} \colon B(X_0r) \subset \mathbb{R}^n \rightarrow \mathbb{R}^m \; g \colon B(X_0r) \rightarrow \mathbb{R} \quad \lim_{\textcolor{red}{X} \rightarrow X_0} \frac{\|\vec{f}(X)\|}{|g(X)|} = 0$$

$$\vec{f} = (f_1 \dots f_m)^T \qquad 1 \leq i \leq m$$

$$\vec{f}(X) = \vec{o}(|g(X)|) = |g(X)|\vec{o}(1) \; (X \rightarrow X_0)$$

$$\textcolor{red}{f}_i(X) = \textcolor{red}{o}(|g(X)|) \; (X \rightarrow X_0)$$

2
$$X_0 = (x_1^{(0)} \dots x_n^{(0)}) \in \mathbb{R}^n \quad r > 0 \quad \vec{f}: B(X_0 r) \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$A: \mathbb{R}^n \rightarrow \mathbb{R}^m \quad X \rightarrow X_0$$

$$\vec{f}(X) - \vec{f}(X_0) = A(X - X_0) + \vec{o}(\|X - X_0\|)$$

$\vec{f}(X_0)$
 A
 $\mathrm{d}\vec{f}(X_0)$
 $\vec{f}(X_0)$
 A
 $\vec{f}(X_0)$
 $J_{\vec{f}}(X_0)$
 $\vec{f}(X_0)$
Jacobi

$$f_i(X_0) = \frac{\partial f_i}{\partial x_j}(X_0) \, \mathrm{d}\vec{f}(X_0) = \sum_j \frac{\partial f_i}{\partial x_j}(X_0) \, \mathrm{d}x_j$$

$$\mathrm{d}\vec{f}(X_0) = \begin{pmatrix} \mathrm{d}f_1(X_0) \\ \vdots \\ \mathrm{d}f_m(X_0) \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n \frac{\partial f_1}{\partial x_j}(X_0) \, \mathrm{d}x_j \\ \vdots \\ \sum_{j=1}^n \frac{\partial f_m}{\partial x_j}(X_0) \, \mathrm{d}x_j \end{pmatrix}$$

$$J_{\vec{f}}(X_0) = \left(\frac{\partial f_i}{\partial x_j}(X_0)\right)_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$$

$$\mathrm{d}X \; \mathrm{d}\vec{f}(X_0) = J_{\vec{f}}(X_0) \, \mathrm{d}X$$

$$J_{\vec{f}}(X_0) \; \frac{\partial (f_1 \dots f_m)}{\partial (x_1 \dots x_n)}(X_0) \; \frac{\partial (f_1 \dots f_m)}{\partial (x_1 \dots x_n)}\Big|_{X_0}$$

$$m = n$$

$$\mathrm{d}\vec{f}(X_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(X_0) & \cdots & \frac{\partial f_1}{\partial x_n}(X_0) \\ \vdots & \cdots & \vdots \\ \frac{\partial f_m}{\partial x_1}(X_0) & \cdots & \frac{\partial f_m}{\partial x_n}(X_0) \end{pmatrix} \begin{pmatrix} \mathrm{d}x_1 \\ \vdots \\ \mathrm{d}x_n \end{pmatrix}$$

Jacobi

$$\frac{D(f_1 \dots f_m)}{D(x_1 \dots x_n)}(X_0)$$

$$\frac{D(f_1 \dots f_m)}{D(x_1 \dots x_n)}\Big|_{X_0}$$

$$\textcolor{blue}{1} \quad \forall (xyz) \in \mathbb{R}^3$$

$$\textcolor{blue}{:} \quad \textcolor{red}{F(xyz) = 0} \quad F \quad \mathcal{C}^{(1)} \quad \begin{matrix} z = f(xy)? \\ F(000) = 0 \end{matrix} \quad \begin{matrix} (00) \\ \frac{\partial F}{\partial z} = xy + e^{x+y+z} \end{matrix} \quad \frac{\partial F}{\partial z}(000) = 1 \neq 0 \quad F(xyz) = 0 \quad F(xyz) = x(1 + yz) + e^{x+y+z} - 1 \quad z = f(xy)$$

$$\begin{aligned}\frac{\partial f}{\partial x}(00) &= -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}\bigg|_{(000)} \\ &= -\frac{1 + yz + e^{x+y+z}}{xy + e^{x+y+z}}\bigg|_{(000)} = -2\end{aligned}$$

$$\begin{aligned}\frac{\partial f}{\partial y}(00) &= -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}\bigg|_{(000)} \\ &= -\frac{xz + e^{x+y+z}}{xy + e^{x+y+z}}\bigg|_{(000)} = -1\end{aligned}$$

$$\begin{aligned} & \textcolor{blue}{2} \quad F \mathcal{C}^{(2)} \qquad F(xyz) = 0 \qquad z = f(xy) \mathcal{C}^{(2)} \quad \frac{\partial^2 z}{\partial y \partial x} \\ & \textcolor{blue}{:} \quad u = \frac{\partial F}{\partial z}(xyz(xy)) \neq 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y} \Big(- \frac{\frac{\partial F}{\partial x}(xyz\textcolor{red}{xy})}{\frac{\partial F}{\partial z}(xyz(xy))} \Big) \\ &= - \frac{1}{u^2} \Big[\frac{\partial}{\partial y} \Big(\frac{\partial F}{\partial x}(xyz\textcolor{red}{xy}) \Big) \frac{\partial F}{\partial z}(xyz(xy)) \\ &\qquad - \frac{\partial F}{\partial x}(xyz(xy)) \frac{\partial}{\partial y} \Big(\frac{\partial F}{\partial z}(xyz(xy)) \Big) \Big] \end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 z}{\partial y \partial x} &= -\frac{1}{u^2} \left[\frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} (xyz(xy)) \right) \frac{\partial F}{\partial z} - \frac{\partial F}{\partial x} \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z} (xyz(xy)) \right) \right] \\
&= -\frac{1}{u^2} \left[\left(\frac{\partial^2 F}{\partial y \partial x} + \frac{\partial^2 F}{\partial z \partial x} \frac{\partial z}{\partial y} \right) \frac{\partial F}{\partial z} - \frac{\partial F}{\partial x} \left(\frac{\partial^2 F}{\partial y \partial z} + \frac{\partial^2 F}{\partial z^2} \frac{\partial z}{\partial y} \right) \right] \\
&= -\frac{1}{u^2} \left[\left[\frac{\partial^2 F}{\partial y \partial x} + \frac{\partial^2 F}{\partial z \partial x} \left(-\frac{\frac{\partial F}{\partial y}}{u} \right) \right] \frac{\partial F}{\partial z} - \frac{\partial F}{\partial x} \left[\frac{\partial^2 F}{\partial y \partial z} + \frac{\partial^2 F}{\partial z^2} \left(-\frac{\frac{\partial F}{\partial y}}{u} \right) \right] \right] \\
&= -\frac{1}{u^3} \left[\left(\frac{\partial F}{\partial z} \right)^2 \frac{\partial^2 F}{\partial y \partial x} - \frac{\partial F \partial F}{\partial y} \frac{\partial^2 F}{\partial z \partial z \partial x} - \frac{\partial F \partial F}{\partial x} \frac{\partial^2 F}{\partial z \partial y \partial z} + \frac{\partial F \partial F \partial^2 F}{\partial x \partial y \partial z^2} \right] \\
&=
\end{aligned}$$

$$= -\frac{\left(\frac{\partial F}{\partial z}\right)^2 \frac{\partial^2 F}{\partial y \partial x} - \frac{\partial F \partial F}{\partial y \partial z} \frac{\partial^2 F}{\partial z \partial x} - \frac{\partial F \partial F}{\partial x \partial z} \frac{\partial^2 F}{\partial y \partial z} + \frac{\partial F \partial F}{\partial x \partial y} \frac{\partial^2 F}{\partial z^2}}{\left(\frac{\partial F}{\partial z}\right)^3}$$

